

Quiz-3

Name: _____ Roll: _____ Date: 24/11/2025

Instructions

1. You are given 60 minutes to answer 4 questions. Please read all questions carefully before attempting.
2. There are 2 MCQ, each having four options, with only one correct answer.
3. There are 2 subjective questions. Answers should be similar to lectures.

All The Best!

1. Consider A to be $n \times d$ dimensional matrix and S be a $m \times n$ dimensional matrix such that for every $z \in R^n$ that are in the column space of A the following holds with a high probability, $(1 - 2\varepsilon)\|z\|_2^2 \leq \|Sz\|_2^2 \leq (1 + 2\varepsilon)\|z\|_2^2$. Let $n > d$ and $\varepsilon \in (0, 1/2)$. What is the smallest required size of the ε -net to satisfy the above condition. **(+2/-1 marks)**

☐ $\left(\frac{\varepsilon}{2}\right)^d$	☐ $\left(\frac{\varepsilon}{2}\right)^{-n}$	☒ $\left(\frac{\varepsilon}{2}\right)^{-d}$	☐ $\left(\frac{\varepsilon}{2}\right)^n$
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2. Let, (P, ω) be a weighted dataset where $\omega: P \rightarrow R_{>0}$ be the weight function on the dataset P . Let $f: P \times \omega \times X \rightarrow R_{\geq 0}$ be a function needs to be optimized over some set X . Let, (P, v) be a weighted set, where $v: P \rightarrow R_{\geq 0}$ be the weight function on the same dataset P .

- a. The set, (P, v) is a weighted subset because, **(+2/-1 marks)**

☒ $ \{p p \in P, v(p) > 0\} < P $
☐ $ \{p p \in P, v(p) \geq 0\} < P $
☐ $ \{p p \in P, \omega(p) > 0\} < P $
☐ $ \{p p \in P, \omega(p) \geq 0\} < P $

- b. The weighted subset (P, v) , is a strong (ε, δ) approximation of (P, ω) for the function f if, $(1 - \varepsilon)f(P, \omega, x) \leq f(P, v, x) \leq (1 + \varepsilon)f(P, \omega, x)$ with at least $1 - \delta$ probability for, **(+2/-1 marks)**

☒ every $x \in X$	☐ $\bar{x} \in X$, where $\bar{x} \in \operatorname{argmin}_{x \in X} f(P, \omega, x)$	☐ some $x \in X$	☐ $\tilde{x} \in X$, where $\tilde{x} \in \operatorname{argmin}_{x \in X} f(P, v, x)$
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- c. For $\varepsilon > 1$, which of the following is true for strong (ε, δ) approximation defined above. **(+2/-1 marks)**

☐ Upper bound is trivial	☒ Lower bound is trivial	☐ Both bounds are non trivial	☐ None of these.
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3. Let A be a $n \times m$ dimensional matrix such that $\mathbf{rank}(A) = k < \min\{n, m\}$. Compute an interpolative low rank approximation of A that consists of the rows of the A . The algorithm must return C and W such that, W is a subset of the rows of A and the following guarantee holds. Hint: Use projection matrix.

$$\|A - CW\|_F^2 \leq \frac{\|A\|_F^2}{3}$$

- a. Design an algorithm that computes C and W . The required dimensions of the C and W must be the initialization parameters of the algorithm. **(5 marks)**
 - b. If P is projection matrix, used in your algorithm then show that $PP = P$. **(2 marks)**
4. Let A be a $n \times m$ dimensional matrix representing n datapoints in m dimensional space. Let b be an n dimensional vector representing the responses of n datapoints. Consider a fixed vector $x \in R^m$, for which the loss is $\|Ax - b\|_2^2$. Answer following with respect to the vector.
 - a. Design a sampling algorithm for the problem. **(5 marks)**
 - b. Show that your sampling method is unbiased. **(3 marks)**
 - c. Ensure that your sampling method guarantees the smallest possible variance for the problem. **(7 marks)**

Answer

3.a) For two matrices P and Q with appropriate dimensions we know that, $\|PQ - CW\|_F^2 \leq \frac{\|P\|_F^2 \|Q\|_F^2}{s}$, where C and W has s columns and rows respectively. The matrices C and W consists of s weighted of columns of P and corresponding rows of Q .

For our problem we have $A \in \mathbb{R}^{n \times m}$ such that $\text{rank}(A) = k < \min n, m$. We will define a projection P matrix from the columns space of A such that $PA = A$. Next, we define a distribution on the rows of A and based on it, sample m columns and corresponding rows from P and A respectively. Since, P is a projection matrix of rank k hence $\|P\|_F^2 = k$. Therefore we set $m = 3k$, to get the desired bound.

Pseudo Code

Input: $A, m = 3k$

Output: C, W

$R \leftarrow$ Randomly sample of k independent columns of A

$P \leftarrow R(R^T R)^{-1} R^T$

Forevery $i \in [n]$

$q_i \leftarrow \frac{\|A_{i:}\|_2^2}{\|A\|_F^2}$ // $A_{i:}$ is the i^{th} row of A .

Based on distribution $\{q_i\}_{i=1}^n$, randomly sample $T \subseteq [n]$ such that $|T| = m$

$i = 0$

While T is not empty

select $j \in T$

$i = i + 1$ // i points to i^{th} column of C and i^{th} row of W .

$C_{:i} \leftarrow \frac{P_{:j}}{\sqrt{q_j}}$ // $P_{:j}$ is the j^{th} column of P .

$W_{i:} \leftarrow \frac{A_{j:}}{\sqrt{q_j}}$

Remove j from T

b) Recall, $P \leftarrow R(R^T R)^{-1} R^T$.

Hence, $PP = R(R^T R)^{-1} R^T R (R^T R)^{-1} R^T = R(R^T R)^{-1} R^T = P$.

Note that by definition, R is a rank k matrix. Hence, $(R^T R)$ is a rank k matrix of size $k \times k$. Therefore, $(R^T R)^{-1} R^T R = I$. This is used above to prove $PP = P$.

Answer

4.a)

Input: A, b, m

Output: A, b, v

For every $i \in [n]$, set $v(i) = 0$

For every $i \in [n]$

$$p_i = \frac{(A_i \cdot x - b_i)^2}{\|Ax - b\|_2^2} \quad // A_i \text{ is the } i^{\text{th}} \text{ row of } A \text{ and } b_i \text{ is its corresponding response.}$$

Based on distribution $\{p_i\}_{i=1}^n$, randomly sample $Q \subseteq [n]$ such that $|Q| = m$.

For every $j \in Q$

$$\text{Set } v(j) = v(j) + \frac{1}{m \cdot p_j} \quad // \text{ For every selected index, assign appropriate weights.}$$

b) Notice, that by definition $\{p_i\}_{i=1}^n$ is a distribution. Now, in the above algorithm every $j \in Q$ is a sample from a random variable

$$r \text{ whose distribution is } r = \begin{cases} \frac{(A_1 \cdot x - b_1)^2}{m \cdot p_1} & \text{with a probability } p_1 \\ \vdots & \\ \frac{(A_n \cdot x - b_n)^2}{m \cdot p_n} & \text{with a probability } p_n \end{cases} . \text{ Now, } r_1, r_2, \dots, r_m \text{ are } m \text{ iid samples from this distribution.}$$

Let, $R = \sum_{i=1}^m r_i$. Now compute $E[R]$ it should come out to be $\|Ax - b\|_2^2$.

c) Compute, $\text{Var}[R]$, it should be the smallest possible variance for any random variable.